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Fuel 278 (2020) 118242

Contents lists available at ScienceDirect

Fuel

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Effect of TMP-based-cottonseed oil-biolubricant blends on tribological behavior of cylinder liner-piston ring combinations

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ARTICLE INFO

Keywords:
Cotton-biolubricant
COF
Wear
Engine cylinder-piston ring

ABSTRACT

Cottonseed oil-based biolubricant was synthesized by the TMP-based transesterification process. 10–50% by volume blends of TMP-based cotton-biolubricant and SAE-40 were prepared and tested on the high-frequency-reciprocating-rig with engine cylinder-liner and piston-ring combination to investigate their tribology. While tribological characteristics were also evaluated by four-ball tribo-testers at high constant load of 785 N. 10% addition of cotton-biolubricant showed the lowest friction and wear as compared to SAE-40 but > 10% volume of cotton biolubricant in blend increased the wear and friction considerably as tested by both HFRR and four-ball. Hence, 10% addition of TMP-cotton-biolubricant can be utilized as an energy-saving lubricant additive to partially reduce the dependency on petroleum-based lubricant for automotive engine application.

1. Introduction

Most of the mineral-based lubricants are available in the global market but very few are synthesized form synthetic and vegetable oils. Prices of petroleum products are increasing due to the rapid depletion of reservoirs. Moreover, petroleum-lubricant are non-biodegradable and toxic which creates health, water, and environmental issues [1].

So, there is a need to increase the synthesis of lubricants from bio-sources like vegetable oils or to find other alternatives. Bio-based lubricants are superior to mineral lubricants because they are biodegradable, non-toxic renewable, and their spillage is not harmful to the environment [2,3].

Some researchers showed that vegetable oils showed better lubricity than mineral and synthetic lubricants because they have polar ester groups along with long-chain fatty acids [4–7]. The polar carboxyl group of fatty acids adhere to the metallic surface and forms an effective thin lubricating film to decrease friction and wear by minimizing metal to metal contact [8,9]. Reeves et al used vegetable oils as lubricants and found that long-chain fatty acids (with C18 and above C-atoms) show good lubricity and anti-wear properties [8]. Ertugrul and Filiz's research showed that a very small amount of cottonseed oil can be used as additives in commercial lubricants to decrease friction [9].

But vegetable oils show poor corrosion protection, poor cold flow properties, low oxidative, and thermal stability because of unsaturation in fatty acids composition. The oxidative degradation produces insoluble deposits that increase viscosity and acidity of oil [10,11].

Oxidation of esters also produces fatty acids like propionic acid, formic acid, acetic acid, and caproic acid that are very corrosive [12].

Although unsaturation in long-chain fatty acids in vegetable oils is good for better lubricity and clod flow properties but decreases the oxidative and thermal stability [13]. Fatty acid's chains of bio-lubricants are very sensitive to temperature and high temperature causes their oxidation that produces free fatty acids. These free fatty acids formed a metallic soap layer between sliding surfaces of metal and effectively reduce the COF but this layer has low strength against shear. For example, the metallic soap of the iron stearate layer is formed by the reaction between iron (from plates) and free fatty acids of stearic acid C18:0 from bio-lubricant [14]. Moreover, oxides of metals also

Abbreviations: CSO, Cottonseed oil; CME, Cotton methyl ester; CBL, Cotton biolubricant; CBL-0, Commercial lubricant SAE-40; FA, Fatty acids; TMP, Trimethylolpropane; COF, Coefficient of friction; Avg.COF, Average Coefficient of friction; WSD, Wear scar diameter; HFRR, High-frequency reciprocating rig; SEM, Scanning electron microscope; EDX, Energy-dispersive X-ray spectroscopy; ASTM, American Society for Testing and Materials

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https://doi.org/10.1016/j.fuel.2020.118242
Received 15 January 2020; Received in revised form 21 May 2020; Accepted 26 May 2020
Available online 18 June 2020
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formed on the metal surfaces by the chemical attack of polar FA and act as metallic soap film. These metallic soap layers have low resistance towards share so formed and removed continuously by sliding motion, thus producing nonreactive detergents that increase the wear [5,15]. Saturated and high concentration of oleic acid (monounsaturated fatty acids – C18:1) have the ability to develop effective FA-monolayer that reduces metal to metal contact and minimize frictional force [16].

Chemical modification of vegetable oils having high oleic fatty acids can improve lubricity, wear, cold flow properties, and thermos-oxidative stabilities simultaneously. The transesterification process can synthesize potential TMP-based bio-lubricant for high pressure and high-temperature applications [10,13,16]. Zulkifli, N.W.M et al showed that some TMP-based-biolubricants have poorer lubricity and wear [17] than ordinary mineral lubricant but the addition of TMP-based-biolubricants as additives in ordinary/commercial lubricant can significantly improve its tribological properties [17,18]. There is also the need of finding cheap feedstock for bio-lubricant synthesis and cottonseed oil is considered a potential source of feedstock obtained from indigenous crops in many countries. Among edible oils, cottonseed oil is less preferred so it can be easily used for fuels or biolubricants synthesis [19]. Cottonseed oil has 22–26% of palmitic acid (C16:0), 2–5% of stearic acid (C18:0), 15–22% of oleic acid (C18:1), and 49–58% of linoleic acid (C18:2). Cottonseed oil has a good balance of oleic and linoleic acids so it can produce good potential TMP-based-bio-lubricant.

George S. Dodos et al measured the load-carrying capacity of (TMP based) cotton biolubricant in terms of scar diameter [20].

Siraskar Gulab Dattrao tested the esterified (TMP based) cotton biolubricant with 50% commercial lubricant in a 4-stroke SI engine for 100 h to evaluate its engine oil temperature and viscosities only. Based on oil temperatures and viscosities, he concluded that TMP-based cotton biolubricant can be utilized as lubricant without the reduction in engine performance, but additives should be used for high load, high temperature and, for longer period applications [21]. Very limited literature is available on cotton biolubricant as no one explored details wear and frictional characteristics of TMP-based cotton biolubricant as an additive for automotive applications.

Bowden and Tabor found that some plant oils produce good boundary biolubricants with low COF but give a high wear rate [22]. Therefore, there is also the need for exploring detailed research on tribological behavior (lubricity, friction, wear, and surface roughness) of chemically modified cotton biolubricant for automotive applications.

In this study, TMP based biolubricant was synthesized from cottonseed oil by the transesterification process. Different percentages (10–50%) of TMP based cotton bio-lubricant (CBL) as additive were mixed with commercial SAE-40 lubricant (CBL-0) to investigate their detailed lubricity and tribological properties for cylinder liner and piston ring combination by using HFRR and four-ball tribo-tester.

2. Chemicals, materials and experimental methods

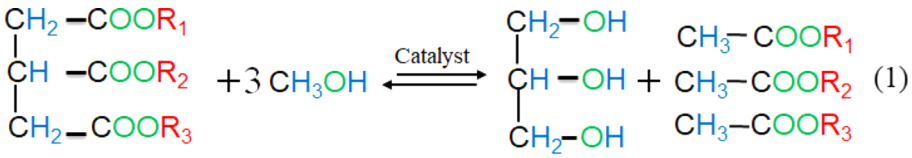
Crude cottonseed oil was bought from the local market of Lahore: Pakistan, Cotton methyl ester (biodiesel) was synthesized from cottonseed oil and methanol by the transesterification process.

Sodium methoxide (30 wt%) solution in methanol (ACROS ORGANICS brand) as a catalyst, trimethylolpropane (TMP) and filter papers were purchased from chemical suppliers in Malaysia. AISI 52100 steel balls (having diameter = 12.7 mm, and hardness = 64–66 RC), grey-cast iron plates, and Riken: piston ring for tribo-tester, Commercial lubricant with the following brand

'AEROIL SAE-40 API-5F/CC (Engine oil): USA Technology (having anti-corrosion, anti-wear, anti-foaming, antioxidants properties) was purchased from local automobile workshop of Kuala Lumpur, Malaysia.

2.1. Cotton-seed oil-based biolubricant synthesis

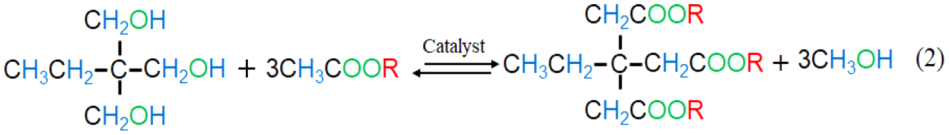
The synthesis of Cotton biolubricant was completed in two steps. In the first step, cotton methyl ester (biodiesel) was produced by reacting triglyceride of cottonseed oil with methanol with 1:6 M ratio at 64 °C and 800 rpm stirring speed in the presence of 1 wt% sodium methoxide catalyst, as shown in equation.1. Glycerol is removed from CME by sedimentation in separating funnel. Purification of CME was performed by heating it 80 °C for 30 min followed by washing with 40 °C hot water several times and then by rotary evaporator.



(1)

Triglyceride of CSO + Methanol Glycerol + CME (Biodiesel)

In the 2nd step, the reaction of the cotton methyl ester with TMP was carried out in a 500 ml 3-neck flask to produce cotton biolubricant in the presence of 0.8 wt% sodium methoxide catalyst as mentioned by equation (2).



(2)

A condenser and rotary vacuum pump were also attached to the flask as the experimental setup is presented in Fig. 1. Vacuum pump applied vacuum and removed methanol from mixture continuously as described in the previously published research paper [17].

The mean molecular weight of the biodiesel (CME) was determined by its fatty acid's compositions, which helps in calculating weights according to the CME: TMP molar ratio as 4:1.

For this process, each time 120 g of cotton methyl ester (CME) was poured into the flask that was dipped in a hot silicon oil bath, and a thermometer is used to measure the temperature. CME was heated up to the 144 °C temperature than the prescribed amount of TMP (with a constant molar ratio as CME: TMP = 4:1) was also poured into the flask. A magnetic stirrer with a constant speed of 1000 rpm was set to mix the mixture thoroughly. Initially, the temperature drops down but when the required 144 °C temperature was attained then 0.8 wt% concentration of sodium methoxide catalyst was added in the reaction flask. The vacuum pump was turned on to apply 25 mbar pressure

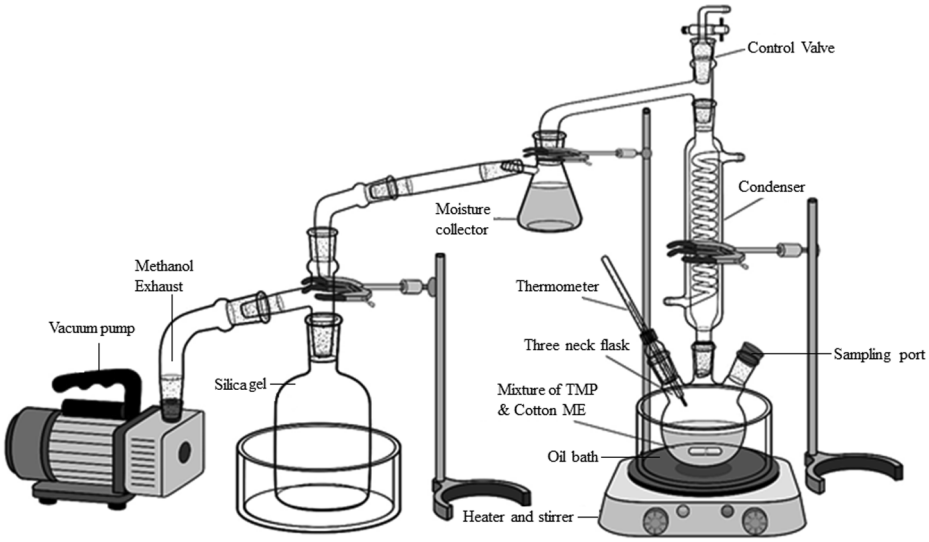


Fig. 1. Experimental setup for transesterification of CME with TMP.

gradually so that no spillage of the reactant mixture occurred. After 10 h of time, the reaction was stopped. Filteration mixture was performed to remove traces of soap and catalyst by using filter paper. Fatty soap was formed due to the presences of Na^+ ions in homogeneous sodium methoxide catalyst.

Pure cotton biolubricant was obtained after separating unreacted CME from the filtered mixture by fractionation via vacuum distillation with temperatures between 150 and 190 °C under very low pressure of about 1–4 mbar.

2.2. Lubricant and bio-lubricant sample preparation

To investigate the physiochemical and detailed tribological properties of TMP based cotton-biolubricant (CBL) as additives, 10–50% various blends of cotton bio-lubricant were prepared with SAE-40 commercial lubricant. While SAE-40 was selected as reference lubricant for comparison purposes. A homogeneous mixture of all samples was prepared by magnetic stirrer at 1100 rpm for 2 hr.

2.3. Physiochemical properties

Density at 15 °C and viscosities at 40 °C & 100 °C were estimated by automatic stabinger viscometer SVM 3000 (Brand: Anton Paar) according to ASTM-D7042

2.4. Experimental setup of friction and wear testing

It is difficult to estimate and develop a statistical correlation between engine performance parameters and tribological characteristics of lubricants in real running engines as it involves high costs for modification and a large quantity of lubricant samples. To avoid such complexity, tribological characterization can be successfully achieved by different laboratory-scale tribo-testers (like four-ball, pin-on-disk, and HFRR) under controlled conditions with standard procedures and give better accuracy [23–25]. These tribo-testers work according to ASTM-standards and also provide effective testing methods correlated with real running engine conditions.

HFRR can be used for different kinds of contact geometries according to the application to detect lubricating film formation for various oils, lubricant's blends and additives in lubricants. It also provides information about the nature of wear at various sliding speeds over small sliding distance (i.e 2–10 mm stroke length) [26,27].

Four-ball tester also used predominantly to estimate the wear and frictional behavior of different lubricating oils at different conditions.

Table 1
Elemental percentage composition of grey cast iron plates.

Elemental composition of grey-cast iron plates (Same as 4-stroke cylinder liner's material)	
Fe	93.39%
C	3.54%
Si	2.34%
Mn	0.42%
S	0.11%
Cr	0.08%
Cu	0.07%
Ni	0.03%
P	0.012%
Others	0.008%
Plates dimensions: (15 × 15 × 4 mm)	

Moreover, it tells about the load-carrying capacity of lubricants at high loads and extreme pressure as described by [28].

2.4.1. High-frequency reciprocating rig (HFRR) tribo-tester

HFRR-tribo-tester is preferred for measuring wear and friction between the cylinder liner and the piston ring [29]. So, HFRR with piston-ring on cylinder liner geometry was selected here for evaluating tribological behavior in IC engines with a very small amount (10 ml) of bio-lubricant blends.

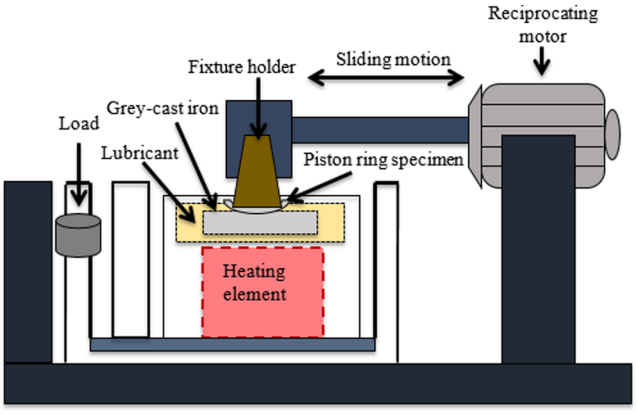


Fig. 2. Schematic view of HFFR wear and friction test-bed.

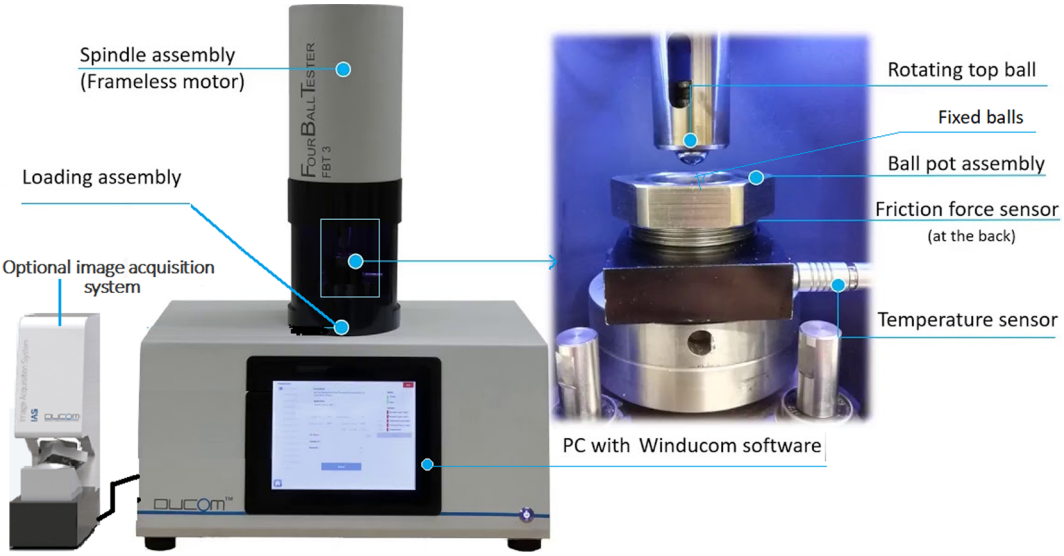


Fig. 3. Four-ball tribo-tester for friction and wear testing at constant load.

Table 2
Physicochemical properties of lubricant's blends.

Physicochemical properties		CBL-0	CBL-10	CBL-20	CBL-30	CBL-40	CBL-50	Accuracy	Standard Method/Equipment
Density (g/cm^3)	@ 15 °C	0.8734	0.8808	0.8848	0.8894	0.8961	0.8991	± 0.0005 g/cm^3	ASTM D7042 Stabinger viscometer
Kinematic viscosity (mm^2/s) or	@ 40 °C	97.706	83.504	68.977	68.767	40.771	31.64	± 0.35%	SVM:3000 Anton Paar
(cSt = Centistokes)	@ 100 °C	18.242	15.909	15.293	17.479	12.348	10.519		
Dynamic viscosity (mPa.s)	@ 40 °C	83.721	72.142	59.876	60.013	35.85	27.918	± 0.01 mPa.s	
	@ 100 °C	14.907	13.1	12.662	14.553	10.36	8.8589		
Viscosity Index	(VI)	207	204.5	235.7	274	314.9	345.7	± 4	

The tribological performances of cylinder liner and piston ring were measured under different bio-lubricant samples (i.e 0%, 10%, 20%, 30%, 40%, 50% blends of CBL with SAE-40) by using high-frequency reciprocating rig (HFRR). For HFRR testing grey cast iron plates (made up of same material as engine cylinder) and standard piston ring of Riken Japan brand (4-stroke engine's piston compression ring made up of cast-iron coated with hard chromium plating) were utilized. Piston ring was cut into equal size pieces according to holder size. The elemental percentage composition of grey cast iron plates is described in Table 1.

The piston ring piece was installed in a fixed holder that reciprocates on the cast iron plates made up of cylinder liner material. Cast iron plate was fixed in the oil tray and fully submerged in the 10 ml lubricant-biolubricant blend samples to be tested for tribology. The required operating temperature of the plate was maintained by heating elements below the oil tray. While, frictional force, wear, and frictional coefficient were measured by electronic transducer against lubrication conditions. All experiments were conducted at 50 N ± 2% load, 50 rh ± 1.5% humidity, 75 °C ± 2% temperature, 6 mm stroke length, reciprocating motion at 10 Hz frequency and 2 Hz frame rate for 1 h (60 min). The schematic view of the HFRR test-bed is described in Fig. 2.

2.4.2. Four-ball tribo-tester

All the cotton biolubricant blends were also tested by using 4-ball tribo tester (DUCOM brand: with ± 1% accuracy) as shown in Fig. 3. The tester consists of an oil cup holder having three stationary balls with a lubricating sample and the fourth ball remains fixed in rotating spindle. AISI 52100 grade 4-steel balls having 12.7 mm diameter and 210 GPa modulus of elasticity with 64–66 Rc hardness were used for each experiment.

Tribological properties were also studied for all CBL- blends (samples) by four-ball tribo-tester at constant high load of 785 N (80 kg). Other working conditions were as temperature = 27 °C, speed = 1770 rpm and time = 10 s. The coefficient of friction was measured by using the following equation.

$$COF\,(\mu) = \frac{\tau\sqrt{6}}{3\mathrm{Lr}}$$

Here, τ represents the frictional torque (Nm), L is applied load (N) and r is the distance from the axis of rotation to the center of contact surface on the lower balls i.e r = 3.67 mm.

While wear scar diameter (WSD) of all the balls was measured by using an optical microscope and their images were taken and analyzed through a well-connected image acquisition system of the optical microscope with DUCOM software of four-ball machine.

2.5. Surface analysis

2.5.1. SEM and EDX analysis

To recognize the lubrication mechanism of lubricant-biolubricant blends on cylinder liner-piston ring conjunction SEM analysis was

Table 3
ISO viscosity grade Standards for lubricating oils [1].

Kinematic viscosity (mm^2/s) or (cSt = Centistokes)	ISO VG32	ISO VG46	ISO VG68
@40	> 28.7	> 41.3	> 61.3
@100	> 4.1	> 4.1	> 4.1
VI	–	> 90	–

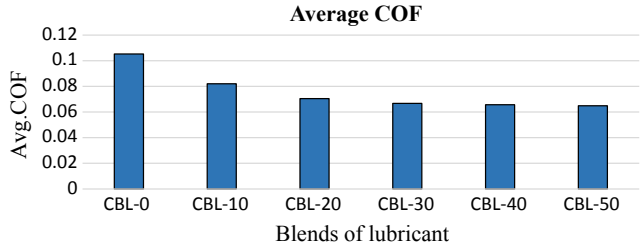


Fig. 4. Variation of avg.COF for all samples under 50 N load and 75 °C by HFRR (50 N/5.1 kg load, 50 rh humidity, 75 °C temperature, 6 mm stroke length, 10 Hz frequency and 2 Hz frame rate for 1 h).

performed on the worn surface of tested plates by projecting a beam of electron on the specimen. As SEM uses conductive type specimen, so specimen plates were mounted on conductive type carbon cement stub so that excess electrons can pass through the specimen. SEM can also produce images at high magnifications.

Energy-dispersive X-ray spectroscopy (EDX/ EDS) was utilized for determining chemical composition or elemental analysis of specimen through X-ray excitation. The unique and specific atomic structure of each element appeared as a peak on the X-ray emission spectrum.

2.5.2. Surface roughness measurement

The surface roughness of cast iron plates was performed at the biomaterial research lab of the dentistry department, UM by using the ALICONA machine. 3D surface-roughness analysis tells about the worn texture and type of wear on the cylinder plates.

3. Results and discussions

3.1. Density and viscosity measurement

The physicochemical properties of all samples are presented in Table 2 and standard values of ISO-VG grade of lubricant are mentioned in Table 3. It can be examined that the density of the samples increases with the increase of cotton-biolubricant proportion in SAE-40 lubricant because of long fatty acid chains. While viscosities at 40 °C & 100 °C decrease due to a high concentration of monounsaturated and poly-unsaturation of fatty acids in cotton biodiesel. The same phenomenon occurs in canola biolubricant derived from biodiesel [30].

Viscosity is the internal resistance against the flow. It gives information regarding the thickness of liquid and how easily it can flow. A liquid having high-density and low kinematic viscosity can flow more quickly [31]. High VI is required for the low rate of change of viscosities with a rise in temperature. Tables 2 and 3 showed that all samples meet the required specification of industrial lubrication oils of IS0 VG 32, 46 and 68 grades. So, high concentration cotton bio-lubricant blends are not good for higher ISO VG-grade applications.

3.2. Tribological characterization by HFRR

3.2.1. Coefficient of friction

For HFRR (reciprocating motion at a relatively low load of 50 N), the resulting COF is primarily governed by the lubricity of the lubricant. The COF of friction reduces as biolubricant concentration increases in SAE-40 blend up to 30% but after that, it is almost constant as presented in Fig. 4. This decrease in avg.COF happens because polar heads of long-chain saturated fatty acids in CBL have a higher tendency to adsorb and react with metallic surfaces, resulting in boundary lubrication effects. Moreover, high concentration of oleic acid (C18:1) also develops an effective FA-monolayer between the sliding surfaces. It is well known that strong surface attraction of the lubricating film and enough cohesive interaction among cotton biolubricant molecules can effectively reduce friction and wear between interacting surfaces [8,24].

Avg.COF of all samples is within the boundary lubrication regime as COF range 0.001–0.2 [7].

Fig. 5 showed the variation of COF against sliding time during run-in-period and steady-state. During the run-in-period, COF remained unstable because of change in torque as contacting surfaces were not smooth initially. After some time, contacting surfaces became smoother and steady-state achieved as stated by [24]. The COF is highest for CBL-0 (i.e commercial lubricant SAE-40). Run-in period and COF decreases with the increase of biolubricant concentration in the blend because long-chain saturated and mono-saturated fatty acids developed stable film between the sliding surfaces. It can also be observed that COF for biolubricant percentages from 30% to 50% is almost the same. Researchers also found that TMP- ester have low COF as compared to commercial and ordinary lubricants [15,17] but TMP-esters increases wear [32].

3.2.2. Surface analysis

3.2.2.1. SEM and EDX analysis. SEM images and EDX results of cast iron plates submerged under different kinds of cotton biolubricant blended with SAE-40 are described in Fig. 6. These images showed the texture of worn surfaces due to the wear between cast iron plates and piston rings. There are various types of wear but boundary lubrication regime occurred during these experiments so adhesive, abrasive, corrosive or fatigue type wears were observed [18]

Commercial lubricating oil anti-wear agent comprises 65–80% by mass of base oil, 2–5% by mass of a rare earth compound, 2–5% by mass of liquid sodium polyacrylate, 5–10% by mass of an antioxidant, and 10–20% by mass of a dispersant or metal oxides [33]. The lubricating oil anti-wear agent uses the unique chemical and physical performances of the rare earth compound that greatly improved wear

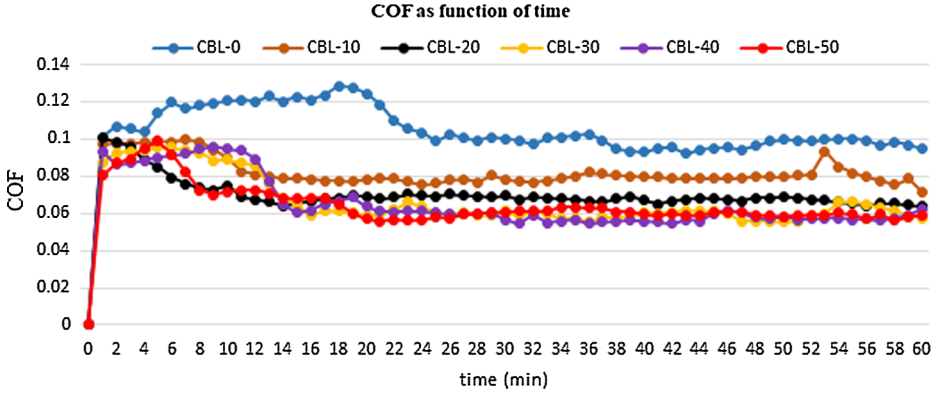
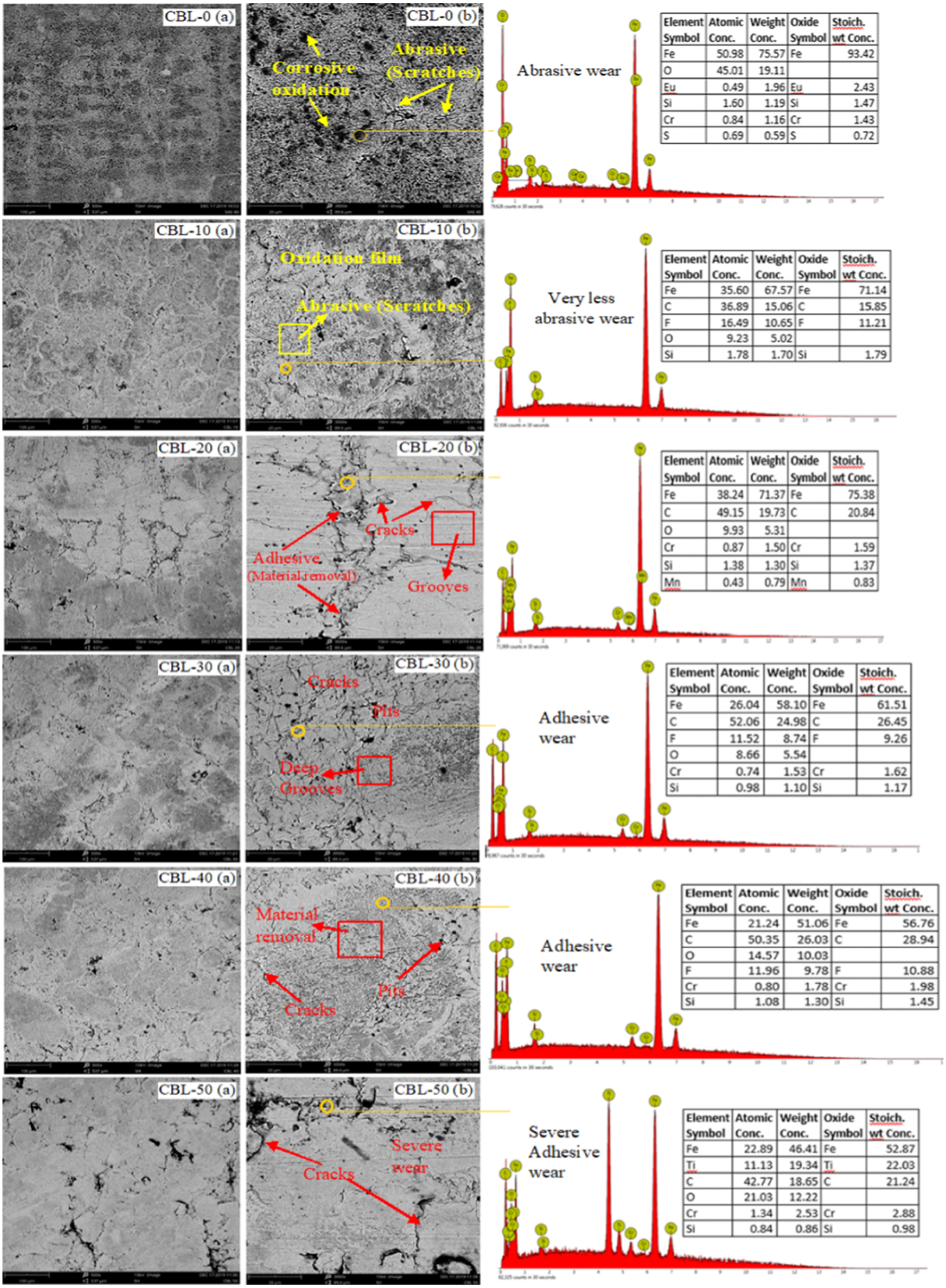


Fig. 5. COF during the run-in period and steady-state period by HFRR.



Element Symbol	Atomic Conc.	Weight Conc.	Oxide Symbol	Stoich. wt Conc.
Fe	50.98	75.57	Fe	93.42
O	45.01	19.11		
Eu	0.49	1.96	Eu	2.43
Si	1.60	1.19	Si	1.47
Cr	0.84	1.16	Cr	1.43
S	0.69	0.59	S	0.72

Element Symbol	Atomic Conc.	Weight Conc.	Oxide Symbol	Stoich. wt Conc.
Fe	35.60	67.57	Fe	71.14
C	36.89	15.06	C	15.85
F	16.49	10.65	F	11.21
O	9.23	5.02		
Si	1.78	1.70	Si	1.79

Element Symbol	Atomic Conc.	Weight Conc.	Oxide Symbol	Stoich. wt Conc.
Fe	38.24	71.37	Fe	75.38
C	49.15	19.73	C	20.84
O	9.93	5.31		
Cr	0.87	1.50	Cr	1.59
Si	1.38	1.30	Si	1.37
Mn	0.43	0.79	Mn	0.83

Element Symbol	Atomic Conc.	Weight Conc.	Oxide Symbol	Stoich. wt Conc.
Fe	26.04	58.10	Fe	61.51
C	52.06	24.98	C	26.45
F	11.52	8.74	F	9.26
O	8.66	5.54		
Cr	0.74	1.53	Cr	1.62
Si	0.98	1.10	Si	1.17

Element Symbol	Atomic Conc.	Weight Conc.	Oxide Symbol	Stoich. wt Conc.
Fe	21.24	51.06	Fe	56.76
C	50.35	26.03	C	28.94
O	14.57	10.03		
F	11.96	9.78	F	10.88
Cr	0.80	1.78	Cr	1.98
Si	1.08	1.30	Si	1.45

Element Symbol	Atomic Conc.	Weight Conc.	Oxide Symbol	Stoich. wt Conc.
Fe	22.89	46.41	Fe	52.87
Ti	11.13	19.34	Ti	22.03
C	42.77	18.65	C	21.24
O	21.03	12.22		
Cr	1.34	2.53	Cr	2.88
Si	0.84	0.86	Si	0.98

Fig. 6. SEM images and EDX spectra of cast iron plates tested with piston ring by HFRR under different cotton biolubricant blends with (a):500× and (b): 3000× magnifications.

Table 4
surface roughness of cast iron plates tested under different blends of bio-lubricants.

Blends of CBL	Surface roughness Ra (μm)		
	Ra (Roughness Average)	Rq	Rz
CBL-0	0.0724	0.0918	0.3778
CBL-10	0.0643	0.0851	0.3534
CBL-20	0.0883	0.1128	0.4715
CBL-30	0.0905	0.1163	0.5007
CBL-40	0.1048	0.1381	0.6002
CBL-50	0.1050	0.1415	0.5744

Table 5
Frictional and wear results by four-ball at a constant high load of 785 N.

Lubricant Blends	COF	WSD (μm)
CBL-0 (SAE-40)	0.22944	2269.8
CBL-10	0.22397	1899.5
CBL-20	0.23914	2203.5
CBL-30	0.26467	2194.33
CBL-40	0.24110	2276.5
CBL-50	0.23202	2424.14

Maximum error WSD = ± 50 μm.
Maximum error COF = ± 0.005.

and extreme pressure resistances of oil. Here EDX results of CBL-0 (SAE-40) showed that it also contains rare earth metal (Eu) to improve wear and extreme pressure resistances of lubricating oil along with other additives that formed a protective film to protect the surface. CBL-0 sample caused corrosive oxidation and abrasive wear on the plate surface.

CBL-10 showed less wear (abrasive wear) because its fatty acids in biolubricant developed an effective protective film between sliding

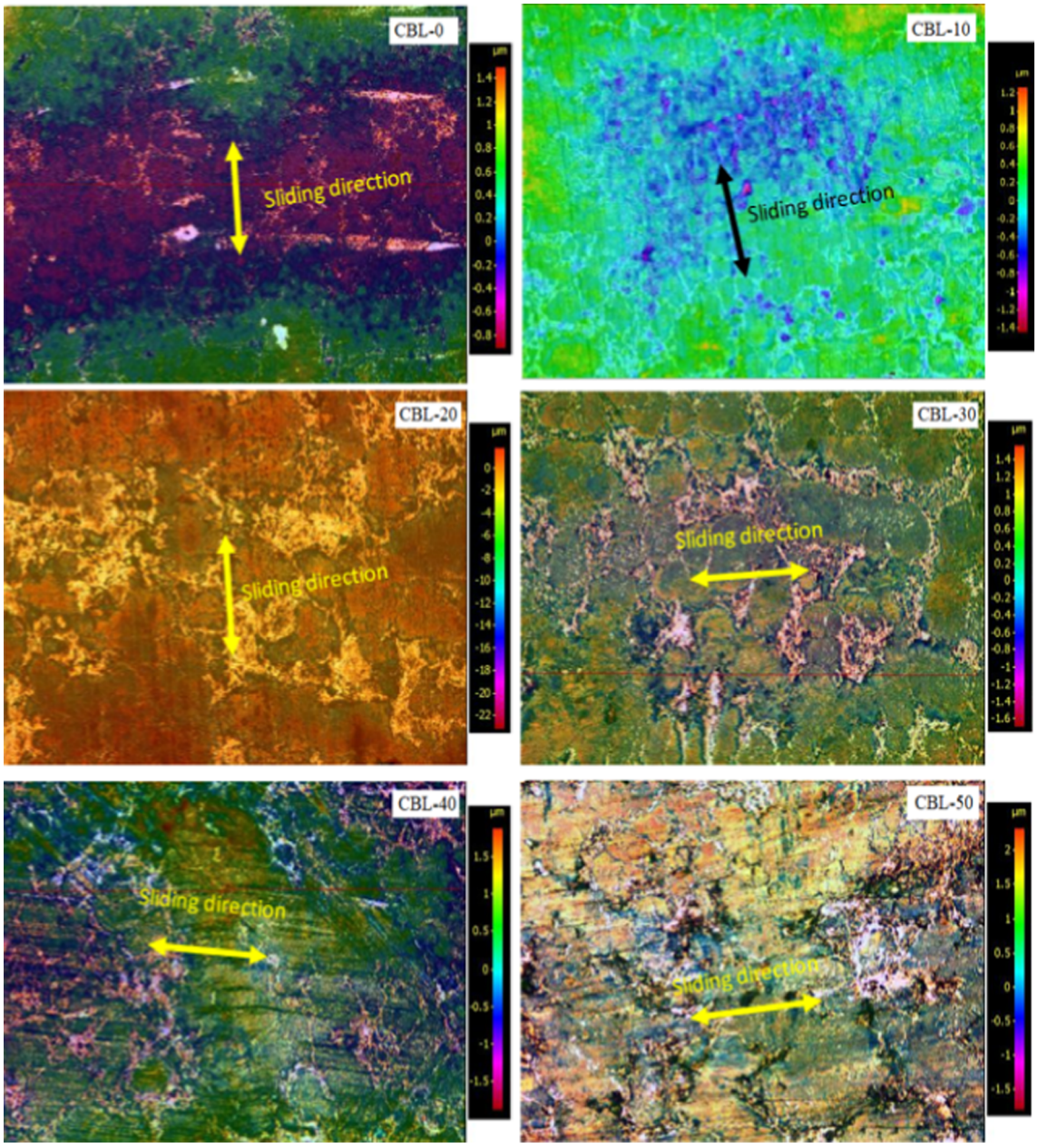


Fig. 7. 3D surface profile (20× magnification) of worn surfaces of cast iron plates tested with piston ring under different cotton biolubricant blends.

surfaces. Wear increased as cotton biolubricant concentration in the commercial lubricant SAE-40 blends increased > 10% because of very low viscosities of blends due to which thin protective film got broken and promote oxidative corrosion. CBL-20, CBL-30, CBL-40, and CBL-50 showed deep grooves, pits, and adhesive wear because material removal occurs. While CBL-40 and CBL-50 showed severe adhesive wear due to oxidative corrosion [15], pits, and cracks.

With the increase of > 10% biolubricant content in the blend caused a decrease in friction but wear increase as shown by severe adhesive wear. This is happening because of the corrosive effects of peroxides and free fatty acids of biolubricant produced during oxidation. Free FA formed metallic soap film by chemical reaction and removed continuously by sliding motion so unreactive detergents increased the wear. The same phenomenon also described by some researchers [5,34].

CBL-0 showed the formation of protective iron-oxide film that decrease wear but lead to the oxidative corrosion resulting increase in scratches as described in Fig. 6.

3.2.2.2. Surface roughness. The 3D surface texture (topography) of worn surfaces of cast iron plates was also calculated by the profilometry test in terms of surface roughness and results are given in Table 4.

This equipment uses the wave-like nature of light to accurately measure distances and evaluate the values of three surface roughness parameters [24] including Ra (roughness average), Rq (root mean square roughness), and Rz (mean roughness depth). Surface roughness tells about the smoothness of any surface.

Less “Ra” value indicates smoothness of surface so it becomes easier for materials to slide together resulting in low COF and low frictional forces [35].

Table 4 showed that the addition of 10% cotton biolubricant in SAE-40 gave minimum surface roughness among all other blends. CBL-10 caused a decrease in Ra, Rq and Rz by 11.2%, 7.3% and 6.5% respectively as compared to pure SAE-40 commercial lubricant.

The 3D surface profiles (topography) of worn surfaces of cast iron plates are also presented in Fig. 7. CBL-0 and CBL-10 showed

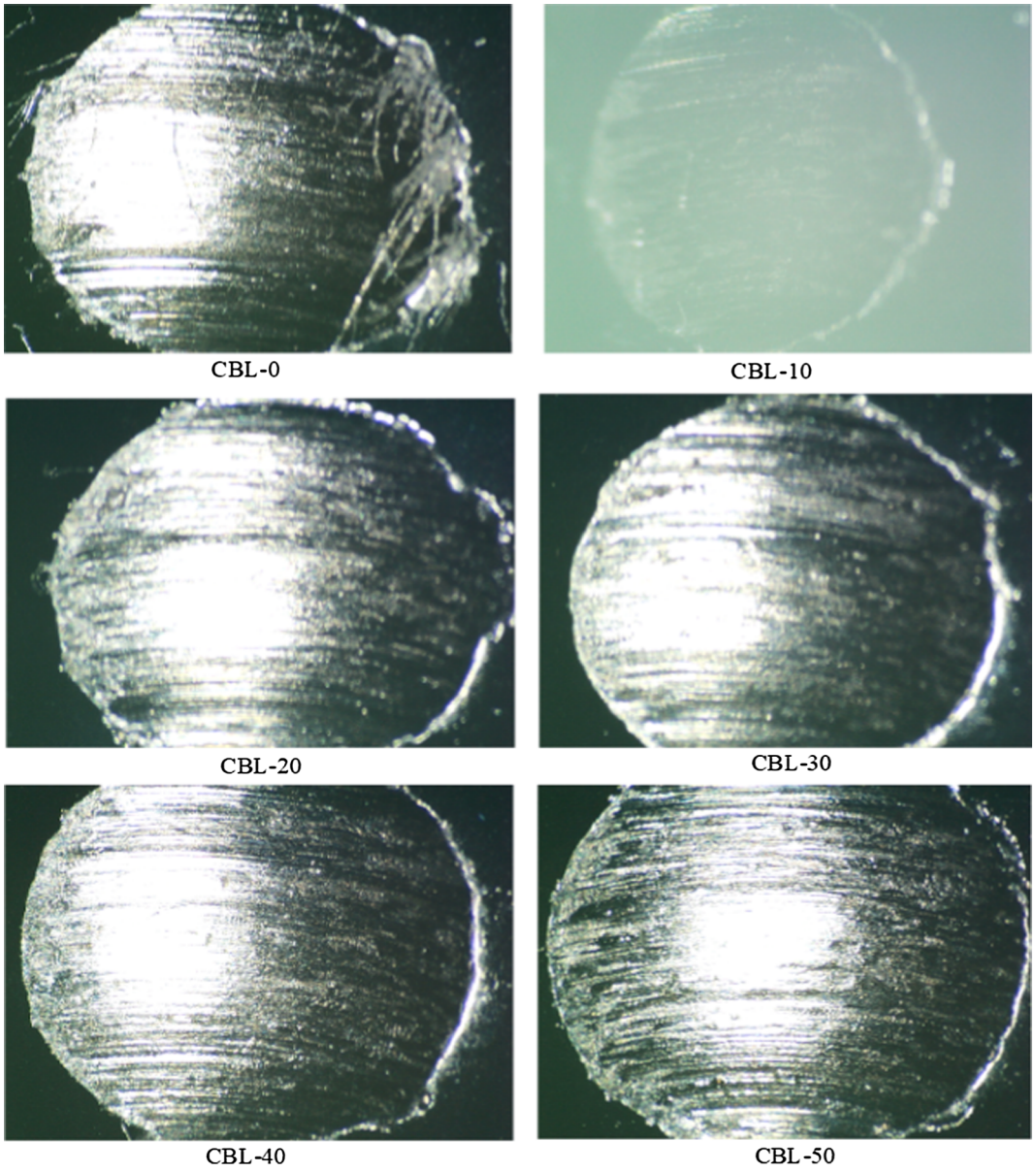


Fig. 8. Microscopic images of balls tested on four-ball image acquisition (300×) with different CBL blends.

comparatively smoother wear patterns. CBL-10 showed minimum surface roughness (Ra = 0.0643 μm, Rq = 0.0851 μm, and Rz = 0.3534 μm) due to the polar fatty acids that make thin but strong oxidative film between the sliding surface to protect them and produced overall smoother surface finishing. But other blends i.e CBL-20, CBL-30, CBL-40, and CBL-50 showed high roughness (Ra, Rq, and Rz) due to corrosive effects of peroxides and free fatty acids that produced during oxidation as described in earlier researches [15].

3.3. Tribological characterization by Four-ball

Tribological analysis of CBL-blends was carried out by four-ball at a constant load of 785 N/(80 kg) and results are presented in Table 5. The COF showed an increase from neat SAE-40 until 30% biolubricant concentration in SAE-40, and then eventually decrease.

The resulting COF is governed by the ability of the lubricant to effectively reduce friction and carry the load during the initial contact (run-in period where friction is the highest) and subsequently form stable tribofilm during the steady-state period. For commercial SAE-40, it produced the lowest COF due to various additives being added during the formulation that can help to carry the heavy load and reduce friction on contact points. When CBL is involved, CBL-10 showed minimum COF and WSD than commercial lubricant SAE-40 and other blends due to the chemical attack of long-chain fatty acids and esters of biolubricans that makes thin, strong and effective lubricating layer (tribofilm) to protect the asperities in the boundary lubrication regime of mating surfaces.

COF showed an increase from 20% to 50% concentration in SAE-40 due to less amount of additives being present in the lubricant, as well as due to lower viscosity of the blend when more CBL is added. Such a phenomenon justified the highest WSD observed for CBL-50 where fatty acid chains are well known to reduce friction but increases the wear due to surface polishing.

Wear scars and microscopic images of worn surfaces of balls tested with different blends of cotton biolubricant are also shown in Fig. 8. Among all blends, CBL-10 also reduced wear and giving small wear scar as visualized by shallow grooves on the ball. The increase of > 10% biolubricant content in the blend caused an increase in friction and wear scar as shown in Fig. 8. This is happening because of the very low viscosities of blends that are not able to withstand the higher loads, thus lubricating layers breakdown results increase in friction and wear. Moreover, this increase in friction and wear at a higher concentration of cotton biolubricant blend was also expected because of the corrosive effects of peroxides and free fatty acids of biolubricant that produced during oxidation. Free FA formed metallic soap film by chemical reaction and removed continuously by sliding motion so unreactive detergents increased the wear and resulting deep grooves. The same phenomenon also described by some researchers [5,34].

> 10% biolubricant content in the blends caused an increase in wear scar because their viscosities decreased which resulted in the thinner weak film between the sliding surfaces. So mating surfaces come closer to each other and increase wear, the same finding also described by other researchers [18,36,37]. CBL-0 also showed less wear scar because CBL-0 is commercial lubricant SAE-40 having anti-wear additives in it.

4. Conclusion

In this study tribological characteristics of cotton biolubricant blends with commercial lubricant (SAE-40) were investigated with cylinder liner-piston ring combination by using HFRR and by using four-ball tribo-testers, following conclusion were drawn

- According to HFRR results, a higher concentration of cotton-biolubricant reduced coefficient (COF) as compared to commercial SAE-40 lubricant. This happens because of fatty acids present in TMP-ester of cotton biolubricant developed an effective lubricating film between the sliding surfaces of cast-iron-plates of cylinder liner and piston ring and increases the lubricity. But the higher concentration of cotton-biolubricant gave higher wear and roughness as observed by SEM and surface roughness analysis.
- Surface roughness (Ra, Rq & Rz) of tested cast-iron-plates also increases with the increase of > 10% biolubricant content in the blends. Only, CBL-10 showed a lower frictional coefficient with less surface roughness.
- Similarly, according to four-ball results at a constant high load of 875 N, CBL-10 showed minimum COF (0.22397) and WSD (1899.5 μm) as compared to SAE-40 and other blends of cotton-biolubricant because it's less viscosity has the ability to developed thinner but strong lubrication layer between mating balls. Hence, CBL-10 can be considered as an energy-saving blend for IC engines.
- CBL-0 (commercial lubricant SAE-40) also showed low friction and wear after CBL-10 because anti-wear additives were present in it. High COF, high WSD, and severe wear were observed in other blends like CBL-20, CBL-30, CBL-40 and CBL-50. This is because a high concentration of biolubricant does not develop strong lubrication films. Moreover, the corrosive effects of peroxides and free fatty acids of biolubricant produced during oxidation also the cause of corrosive and adhesive wear.
- Hence it is concluded that less concentration of TMP-based-cotton-biolubricant as an additive in the commercial lubricant is effective in decreasing friction and wear with good lubricity but a high concentration of CBL is not recommended as it increases the wear. So anti-wear additives should be used with pure cotton biolubricants to get better results.

CRediT authorship contribution statement

M. Gul: Investigation, Methodology, Writing - original draft. **N.W.M. Zulkifli:** Project administration. **H.H. Masjuki:** Supervision. **M.A. Kalam:** Funding acquisition, Resources. **M.A. Mujtaba:** Conceptualization, Data curation. **M.H Harith:** Formal analysis, Software. **A.Z. Syahir:** Validation. **Waqar Ahmed:** Visualization and drawing of Figures. **Abdul Bari Farooq:** Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

Authors would like to thanks all the members of CES, Faculty of Engineering, University of Malaya, Malaysia for their support through the research grant no FP142-2019A under the Fundamental Research Grant Scheme (FRGS) from Ministry of Higher Education, Malaysia and Bahauddin Zakariya University, Multan, Pakistan for the financial support under the Higher Education Commission Project “Faculty Development Program of Bahauddin Zakariya University, Multan (100 Ph.D. Foreign Scholarships) (Prime Minister's Directive), Pakistan” with grant No. PF/18/22-80/Admin/13101.

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DOI: https://doi.org/10.1016/j.fuel.2022.125742

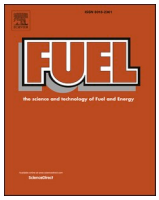
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Corrigendum to "Effect of TMP-based-cottonseed oil-biolubricant blends on tribological behavior of cylinder liner-piston ring combinations" [Fuel 278 (2020) 118242]

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The Authors regret to mention that in the above-published paper an error occurred in the acknowledgment section. The research Grant No FP142-2019A mentioned was incorrect while the correct one is **(FRGS/1/2019/TK03/UM/01/1) – FP142-2019A** under the Fundamental Research Grant Scheme (FRGS) from the Ministry of Higher Education, Malaysia.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

DOI of original article: https://doi.org/10.1016/j.fuel.2020.118242.
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https://doi.org/10.1016/j.fuel.2022.125742
Received 16 August 2022; Accepted 18 August 2022
Available online 31 August 2022
0016-2361/© 2022 Published by Elsevier Ltd.